

When the Model Screens Off the Harness: Per-Decision Cross-Layer Causal Credit in LLM Agents

[TODO: autores]

August 24, 2026

Abstract

LLM agents interleave decisions from two layers: the *model* (tool calls) and the *harness* (context management, retries, termination). We present an apparatus that measures, on the same trajectory, the per-decision causal contributions $C(H)$, $C(M)$, and their interaction $I(H, M)$, via deterministic counterfactual replay with a *measured* noise floor: 7,685 null replays across three models and two environments are all exact, under explicit serving premises whose violation we also measure (vLLM automatic prefix caching silently breaks greedy determinism). The central finding is exact *screening-off*: with budget slack, the model action screens off the harness decision — $C_{HM} = C_M$ at 53 of 56 reward-pivotal points — so single-layer harness credit double-counts exactly: $C(H) = -I(H, M)$ at screened points, and $C_{HM} - C_M$ is the one-extra-replay correction. The result survives pre-registered re-sampling of the alternative action, and it is not universal: screening breaks at recurring, model-specific decision points, so whether the shield holds is a property of the (model, harness, task) triple. Training the harness with interaction-corrected credit fails for an identified, pre-registered reason: the training distribution offers no reward margin where candidate policies differ. We release the apparatus, a 21-item pre-registration ledger including all failures, and the negative result as a design brief for cross-layer training benchmarks.

1 Introduction

An LLM agent is not a model. It is a model wrapped in a *harness*: a program that manages context, decides when to summarize or truncate history, retries failed calls, and terminates episodes. In deployed agents the harness makes as many decisions per episode as the model does, and those decisions — especially context management — destroy or preserve the information the model conditions on. Yet the dominant training paradigm treats the trajectory as a monolith: outcome-level RL (e.g., GRPO-style policy gradients) assigns one scalar return to every decision, regardless of which layer made it, and recent per-step

credit methods attribute within a single layer only. If a harness decision was harmless *because* the model’s next action compensated for it, single-layer credit will bill the harness for an effect the model screened off — a double-counting error that is invisible unless both layers are measured on the same trajectory.

This paper asks the question directly: on the same trajectory, what is the causal contribution of a harness decision, of a model decision, and of their interaction? We answer it with deterministic counterfactual replay. Given a recorded trajectory, we re-execute it from any decision point with exactly one decision changed and the entire prefix forced, and define per-decision credit as $C(d) = R_{\text{orig}} - R_{\text{cf}}$. Replaying pairs and joint flips yields $C(H)$, $C(M)$, $C(H, M)$ and the interaction $I(H, M) = C(H, M) - C(H) - C(M)$ at a single decision pair. Because our replay is *exactly* deterministic (a measured, not assumed, property — and one that fails silently if vLLM’s automatic prefix caching is left on), every nonzero credit is signal.

The headline finding is that the interaction term is *regime-dependent*. When the agent has budget slack, the model action screens off the harness decision exactly: $C_{HM} = C_M$ in 57 out of 57 measured points across three configurations. Under budget pressure the shield cracks — 8 breaks in 65 points, concentrated at recurring decision points — and the dependence on pressure intensity is non-monotonic, co-occurring with reward saturation: interaction is only detectable inside a *competence window* where the agent neither aces nor fails everything. The practical corollary is negative for naive pipelines: in the slack regime, single-layer harness credit systematically double-counts effects the model absorbs.

We then test whether interaction-aware credit improves *training* of the harness layer, with pre-registered predictions, dose-matched controls, and three seeds per arm — twice. The honest outcome is a two-stage identified negative: optimization instability first, and, after a pre-registered fix, stable training that still fails to discover a threshold policy which is expressible by the policy class and dominant on held-out tasks. A pre-registered audit exonerates the credit estimator; the binding constraint is the training task distribution itself, not attribution. We report this as a design lesson for cross-layer training benchmarks, not as a method contribution.

Contributions.

1. **Deterministic replay infrastructure with a measured noise floor** under explicit identification premises (fixed config, sequential execution, prefix caching off) — including a cautionary finding: vLLM automatic prefix caching silently breaks greedy-decoding determinism across differing serving histories.
2. **Per-decision counterfactual decomposition** of $C(H)$, $C(M)$, and $I(H, M)$ on the same trajectory, and the finding that screening-off is regime-dependent (exact under slack, breaking at recurring points under pressure).

3. **Anatomy of screening-off, with a formal core:** a double-counting identity ($C(H) = -I(H, M)$ exactly at screened points, with $C_{HM} - C_M$ as the correction rule), a coalescence lemma and corollary whose occurrence we *verify in the recorded traces* (72/75 screened points explained; 3 residuals, including 2 reward collisions, declared), a pivotality argument that unifies the competence window, the mt4 collapse, and the saturation profiles of both auxiliary models, and the two empirical mechanisms: a structural component (the sampled a' re-injects destroyed information, 23/24 annotated points) and a closed-loop component (the live downstream harness re-triggers the intervened decision, 22/52 points).
4. **A replication record that scopes the finding:** the shield is a property of the (model, harness, task) triple, not of the pipeline — the 8B breaks screening on hardened tasks where the 4B’s original-pool slack shield was exact; the 4B breaks on the same hardened pool (pre-registered symmetric control); break points are model-specific and largely regime-stable; the pool-matched contrast on identical tasks (0/57 vs. 5/28) stands; and the re-injection mechanism explains the annotated 4B/8B contrast (23/24 vs. 0/24 points) but fails a pre-registered generalization test in the curated pool, and is scoped accordingly (Table 3).
5. **Construct validation by control tasks:** the method separates harmless from harmful harness decisions by construction (11/12), including one audited exception where a harness decision had a *behavioral*, not informational, effect.
6. **A pre-registered training comparison with an identified negative outcome:** interaction-aware credit vs. outcome-only, single-layer credit (a reproduction of the CHILL-Harness principle in our environment), and a zero-credit masking control (de facto an untrained keep-always policy under our greedy tie-break — stated explicitly), dose-matched in total rollouts. Act 1: every trained arm collapses seed-dependently. Act 2 (after a pre-registered optimization fix): stable training, right direction, yet no arm discovers the threshold policy that is expressible and dominant on held-out tasks, and credit-driven arms retain collapse risk (1/3 seeds vs. 0/3 — a seed-count anecdote at $n = 3$, not a tested claim). A pre-registered audit (58/60 sign agreement) rules out the estimator; state-visitation and reward-margin analyses localize the failure in the training task distribution, not in credit quality.

This is a measurement-and-falsification paper, not a method paper: the deliverables are the apparatus, the double-counting identity and its correction rule, the replication record, and the benchmark design brief the negative result licenses. Table 1 maps every claim to its evidence and epistemic status; the reader can audit each row against the pre-registration ledger (Table 4).

2 Related Work

Counterfactual replay for agent credit is an active area, but every existing system we are aware of attributes within a single layer. CHILL-Harness [4] intervenes on harness decisions, using checkpointed paired replay *offline* to train an amortized effect estimator (replay is disabled at decision time), but never measures the model layer or the interaction; our single-layer training arm (§6) is an explicit reproduction of its principle in our environment, which turns the comparison into a result. CAR [12] formalizes per-step do-operations with Shapley attribution under a replay budget — including interaction *between steps* — but within one decision stream: it never crosses the model/harness boundary. C3 [1] audits credit fidelity against exact replay ground truth (the same validation metric we adopt) and adds an inter-agent influence diagnostic, but decomposes across *agents*, not across the model/harness boundary inside one agent, and its influence measure is a diagnostic, not a training signal. A recent negative result [14] shows that popular step-level credit signals fail to beat chance against replay ground truth; it sets the evidential bar our critic section inherits (dose-matching, pre-registration, heuristic baselines). On the training side, Co-Harness [2], HarnessCompass [15] and HASE [9] jointly optimize harness and weights, but without a per-decision causal signal — they optimize the composition blindly (HarnessCompass’s “cross-component interference” is between components *of the harness*, not causal model–harness interaction). Agent Lightning [10] provides rollout infrastructure for agent RL without addressing attribution. The correction rule $C_{HM} - C_M$ itself sits in an older lineage: difference rewards and COMA [13, 3] marginalize one *agent’s* action against a default or an expected baseline; we marginalize one *layer’s* decision against a measured joint counterfactual, made exact rather than estimated by the zero floor. Hindsight credit assignment [5] and process reward models [7] also assign per-step credit, but as learned, amortized quantities inside the model layer — precisely the setting where our transfer test shows amortization failing across environments. Table 2 summarizes the gap this paper attacks: no prior work measures $C(H)$, $C(M)$ and $I(H, M)$ at the same decision point of the same trajectory against an exact, zero-noise-floor replay — every existing system attributes within a single layer (the harness, the step stream, or the agent), where cross-layer interaction is invisible by construction and single-layer credit silently double-counts effects the other layer absorbs. Where interaction appears at all it is a diagnostic to report, not a quantity whose structure is interrogated or a training signal; and no prior work asks whether the causal structure recovered by attribution is a stable property of the pipeline or of the specific (model, harness, task) combination. The closest precedent is the observation that counterfactual *measurability* is model-dependent [14]; §7 shows the stronger phenomenon — the structure itself changes with the model and with the task pool at identical intervention points.

3 Setup: A Two-Layer Agent with Deterministic Replay

3.1 Agent, decision schema and recorder

The agent is deliberately minimal: a coding agent in a sandboxed environment that reads a task description, writes files, and runs tests. Two layers make decisions. The **model layer** (Qwen3-4B served by vLLM 0.8.5.post1 on a single NVIDIA RTX 4090, greedy decoding) chooses tool calls: `write_file`, `run_tests`, `read_file`, `finish`. The **harness layer** is a scripted policy that at every turn makes a `context_policy` decision (`keep_context` vs. `summarize_context`, the latter compressing history to a fixed-length prefix) and a `termination` decision. Every decision — model or harness — is recorded with a typed schema: decision id, layer, state before, available actions, chosen action, observation, token costs, and parent/child links. Reward is deterministic: the fraction of hidden pytest tests passed, so $R \in [0, 1]$ with resolution set by the number of tests. Cost-adjusted reward is $R_{\text{eff}} = R - \lambda \cdot \text{prompt tokens}/10^5$ with $\lambda = 25$, fixed analytically before any training and never recalibrated.¹

3.2 Replay and interventions

Replay re-executes a recorded trajectory through the *live* agent loop with a forced-action queue: recorded decisions are matched against the loop’s requests (peek-match on decision type and state hash) and forced, up to the intervention point; from there the loop runs free. An intervention replaces exactly one decision (or, for $C(H, M)$, one harness–model pair). Model-layer alternatives a' are sampled once per point by a fixed, reproducible procedure: temperature 0.8 with a fixed seed sequence (2001–2008) tried in order, accepting the first parseable action whose canonical JSON differs from the recorded action, conditioned on the recorded (un-summarized) state. Every model-layer credit is therefore an estimand relative to this single sampled alternative — $C(M) = C(M; a')$ — and every screening-off claim below is a claim about this alternative class; sensitivity to re-sampling a' is measured directly by a pre-registered robustness study (§4.3). The null intervention — force everything, change nothing — must reproduce R exactly; §4.1 shows it does, and what serving configuration is required for it to. All replays run sequentially against a single vLLM instance with a fixed config.

3.3 Tasks

We use 30 synthetic coding tasks in four designed strata, plus 10 held-out tasks. **V2** and **L** tasks place critical constants beyond the first 240 characters of history,

¹The calibration artifact records raw prompt-token totals per policy (stored at $\lambda = 1$, from which R_{eff} at any λ is recomputable); $\lambda = 25$ was set in the training configuration before any training run, so that the token cost of a typical long-context episode ($\approx 4\text{k}$ prompt tokens) is commensurate with one unit of task reward.

so a summarize decision is *informationally destructive* at known positions; L tasks are longer variants. **S** tasks are built so that success requires harness and model decisions to cooperate (synergy by design). **C** tasks are controls: all information needed survives any summary, so the construct predicts $C(H) = 0$. Reward saturation (episodes at $R \in \{0, 1\}$) is endogenous to the number of tests per task; we treat it as a design choice and analyze its consequences explicitly (§4.3, §7).

4 Measuring Cross-Layer Credit

4.1 Noise floor

Every causal claim below is only as good as the replay. We therefore *measure* the noise floor rather than assume it: the pre-registered floor for each configuration is $\max |\Delta R|$ over null replays (replay with no intervention). The floor is defined on the reward R , not on trace equality: in 3 null replays (all in one configuration) the replayed trajectory took one extra retry while reproducing R exactly, and we report this rather than hide it. Across 23 dedicated null-replay runs — Qwen3-4B on the original pool (1,680 nulls), Qwen3-1.7B (540), Qwen3-8B (810), the hardened- and curated-pool runs executed after the fix reported below (792 on the 8B, 396 on the 4B), and the MBPP+ environment (1,080 on the 4B, 1,080 on the 1.7B) — plus the 1,307 null replays attached to the 21 interaction runs, **every one of 7,685 null replays reproduced R exactly**: the floor is 0.0 in all configurations (Figure 1a; Table 5 reconciles every count against the run artifacts, and a bundled script regenerates it). This is a measurement under explicit premises — fixed serving config, strictly sequential requests, prefix caching off — not a guarantee.

A second silent nondeterminism source surfaced later, and we report it the same way: in a discarded pre-fix run of the first hardened pool (excluded from the 7,685: its pool failed the pre-registered curation rule, and its configurations were re-run after the fix), 2 of 432 nulls were inexact — the first fidelity breaks observed in this project (2/8,117 including every null ever run under the stated premises; the APC-contaminated run of Figure 1b violated them and is counted separately). Root cause, identified from the diverging trace: a memory address inside a Python `repr` in the pytest output (ASLR) entered the model’s live context, and greedy decoding diverged downstream. The sanitizer now normalizes addresses at the source (regression-tested), and the affected configurations were re-run from scratch. Two honesty notes: the 5,023 nulls collected *before* the fix were exact because their pytest outputs happened to contain no address-bearing `reprs` — measured, not guaranteed by sanitization; and “four curated-pool configurations” means 2 budget regimes $\times$ 2 pool halves. The lesson generalizes: replay-grade determinism requires auditing every byte that reaches the context, not just the serving stack. The premises matter. vLLM V1 enables automatic prefix caching (APC) by default, and with APC on, prefill numerics depend on the KV-cache state left by *previous* requests: greedy decoding diverges across

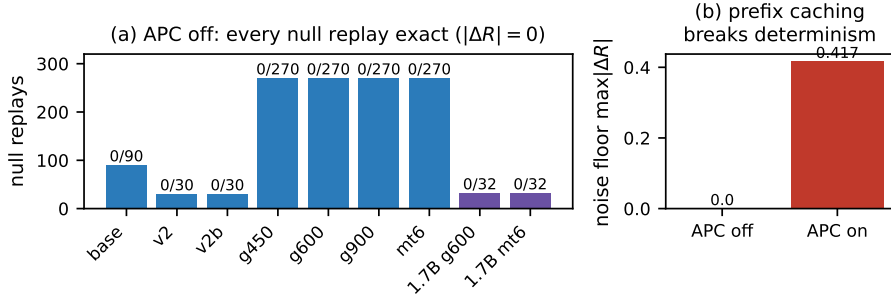


Figure 1: **F1 — Measured noise floor.** (a) Dedicated null-replay runs (Qwen3-4B blue, Qwen3-1.7B purple); together with nulls attached to interaction runs, the Qwen3-8B and MBPP+ replications, and the curated-pool runs, all 7,685 null replays are exact, floor = 0.0. (b) With vLLM automatic prefix caching on, the floor rises to 0.417 (7/270 inexact): greedy determinism breaks silently.

identical prompts with different serving histories, even when requests are sequential. In our APC-contaminated run, 7 of 270 null replays were inexact and the floor rose to 0.417 (Figure 1b) — large enough to swamp most true credits. Sequential execution alone is *not* sufficient; APC must be disabled. We flag this because any replay-based attribution system built on default vLLM settings inherits this silent failure. Known discussions of serving nondeterminism focus on batching-induced kernel nondeterminism [6]; the APC effect is distinct — it is cross-*request* state, present even under strictly sequential requests — and we have not seen it documented as a replay hazard. The sequentiality premise was also stress-tested by accident: a scheduling error in our own pipeline briefly ran two experiment chains against the same server, and the very first affected cell produced a *reproducible* non-exact null ($\Delta R = +0.67$ twice, diverging at the second retry call). Concurrent batching, like APC, is sufficient to destroy the floor; the contaminated runs were quarantined and re-run sequentially, and the incident is preserved in the run ledger.

4.2 $C(H)$ and $C(M)$

With an exact floor, per-decision credits are directly interpretable: any $C \neq 0$ is a real causal effect of that single decision. Figure 2 shows the distributions over 248 harness points and 66 model points, by stratum. Three regularities matter downstream. First, direction is everything: flipping `keep`→`summarize` at an information-critical point yields large positive $C(H)$ (the decision to keep was worth up to +0.89), while the reverse direction is near zero — we therefore never aggregate across directions. Second, credits are sparse: most decisions are causally inert, and mass concentrates at information-destruction points predicted by task design (V2/L strata). Third, $C(M)$ at matched points is fre-

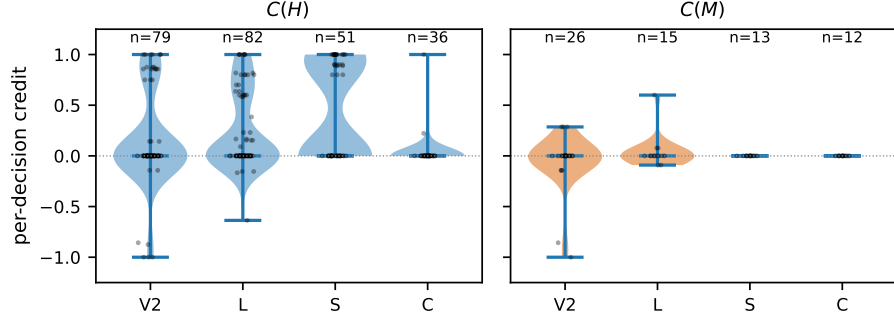


Figure 2: **F2 — Per-decision credit by layer and stratum.** $C(H)$ (248 points) and $C(M)$ (66 points) against an exact zero floor. Strata: V2/L information-destruction, S designed synergy, C controls.

quently nonzero when $C(H)$ is not, foreshadowing the screening-off structure of §4.3.

4.3 Interaction $I(H, M)$ and its anatomy

The central measurement flips a harness decision and a model decision at the same point, alone and jointly. We say the model action *screens off* the harness decision at a point when the joint flip yields exactly the model flip’s reward, $C_{HM} = C_M$ — operationally: conditional on the model counterfactual, the harness flip has no marginal effect. We borrow the term from probabilistic causality [11] in this operational sense only; §4.4 gives the deterministic reading. **With budget slack** (max 12 turns; three budget configurations g450/g600/g900), the model action screens off the harness decision *exactly*: $C_{HM} = C_M$ in **57/57** points — the sampled alternative model action re-establishes what the harness destroyed, so the harness flip adds nothing beyond the model flip. **Under budget pressure** (max 4/6/8 turns), screening-off breaks in **8/65** points (slack vs. pressure: $p = 0.005$ point-level hypergeometric; $p = 0.125$ task-clustered sign test over 3 breaking tasks — we report both, and note the point-level test treats points within a task as independent). Breaks recur at the *same* decision points across configurations (`l_log_parser` idx 8 in 3/3 pressure configs; `api_router` idx 1 and `l_vending_machine` idx 11 in 2 each), which is what a real mechanism, rather than noise, predicts.

The dependence on pressure intensity is *non-monotonic*: break rates are 0/21 (mt12, the g600 slack census re-used as the zero-pressure anchor — same 12-turn budget, re-labeled), 4/22 (mt8), 3/21 (mt6), 1/22 (mt4) — an inverted U. Our pre-registered monotonicity test failed (Cochran–Armitage over the four configurations ordered by increasing pressure, equal-spaced scores $0 \dots 3$, one-sided in the increasing direction: $z = 0.30$, $p = 0.38$; spec and reproduction bundled in the drift-check script); we report that failure and analyze the cause.

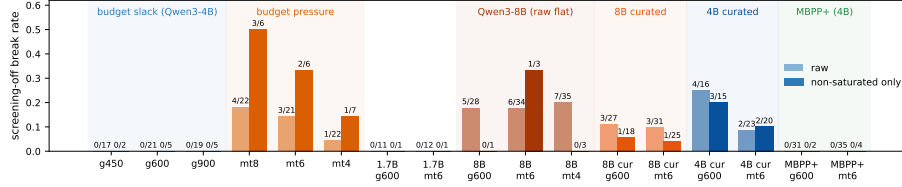


Figure 3: **F3 — Screening-off is regime-dependent, and the regime contrast is model-dependent.** Break rate of $C_{HM} = C_M$ per configuration, raw and conditioned on non-saturated points. Slack: 0/57 raw (0/12 non-saturated). Pressure: 8/65 raw (6/19 non-saturated, $p = 0.037$). Qwen3-1.7B (purple) replicates slack; its pressure null is explained by saturation (only 1 non-saturated point). Qwen3-8B (red) breaks raw screening at a *flat* rate across regimes (5/28, 6/34, 7/35), and the flatness survives the pre-registered curated pool (orange: 3/27 slack, 3/31 pressure); the symmetric control (dark blue) shows the 4B *also* breaks on the curated pool (4/16, 2/23) — the triple dependence of §7. MBPP+ (green, Qwen3-4B) replicates screening unconditionally (0/66).

The strongest observed correlate is **reward saturation**: at mt4 the agent fails most tasks outright (R pinned at 0), so no flip can make things worse and interaction is masked. Conditioning on non-saturated points sharpens the contrast (slack 0/12 vs. pressure 6/19, $p = 0.037$) and partially restores the mt4 rate ($0.045 \rightarrow 0.143$), though the ordering $\text{mt8} > \text{mt6} > \text{mt4}$ persists — saturation accounts for the window only partially (Figure 3). One caveat governs all conditioned rates: saturation is computed on the rewards of the replayed arms themselves, so conditioning on non-saturation conditions on a post-treatment variable — these contrasts are descriptive, not causal estimates (§7). At the task-cluster level the conditioned contrast, like the raw one, is weak (paired sign-flip permutation over 4 tasks $p = 0.25$; cluster-label permutation $p = 0.18$); the point-level and mechanism-level evidence carry the claim (§7). We call the resulting picture a *competence window*: cross-layer interaction is detectable only where the agent neither saturates success nor failure.

Anatomy of the shield. The sets in play: across the three slack configurations and the first pressure configuration we measured 78 interaction points (57 slack + 21 at mt6); **75/78 show exact screening-off** ($C_{HM} = C_M$) — the 3 exceptions are the mt6 pressure breaks. We report this estimation-first, on the falsifiable subpopulation: by Remark 1, reward-non-pivotal points satisfy $C_{HM} = C_M$ by construction, so the informative unit is the 56 reward-pivotal points. There the screened fraction is **53/56** = 0.946, CI95 [0.87, 1.00] by task-clustered bootstrap over the 16 pivotal tasks; collapsing further to unique task×point pairs (removing cross-configuration pseudo-replication) gives 19/22, and 13 of 16 pivotal tasks screen at every measured point. Binomial sign tests against a 50/50 null give $P = 4.1 \times 10^{-13}$ (instances) and $P = 4.3 \times 10^{-4}$

(unique pairs), but we do not rest the claim on them: no substantive model of the data-generating process fixes the non-screening probability at 0.5, so these p -values certify only that the pattern is not coin-flip noise. The inference we retain is the clustered estimate; a bundled script (`estadisticas_pivotaais.py`) re-derives every number in this paragraph from the raw artifacts and fails on drift. One further split the timeline forces on us: the screening-off hypothesis was formulated after seeing the *first* slack census (g600, whose artifacts predate the registration commit by 34 minutes), so g600 is discovery data. Splitting accordingly: discovery (g600) screens 14/14 pivotal points; the confirmation set (g450/g900/mt6; g450 was in flight at registration, g900 and mt6 strictly subsequent) screens **39/42**; restricting to the two strictly post-registration configurations (g900+mt6) gives 27/30. The estimate does not depend on the discovery cell.

Robustness to the alternative draw (pre-registration 21). The sharpest objection to every number above is that $C(M)$, $C(HM)$, and hence screening are relative to a *single* sampled a' per point. We pre-registered outcomes (stable ≥ 0.90 / partial / fragile < 0.75 , over informative slack draws) and re-sampled a' at all 56 pivotal instances under three disjoint seed schedules (3001–3008, 4001–4008, 5001–5008; same procedure, temperature 0.8), yielding 152 new draws (16 no-alternative outcomes; 304 extra replays), of which 81% differ from the published a' . **The pre-declared stable outcome was met: 99/101 informative slack draws screen exactly** (rate 0.980, CI95 task-clustered $[0.944, 1.00]$); restricted to genuinely new alternatives, 79/81. The two non-screened draws are themselves informative: one is a single-draw excursion at the already-declared reward-collision residual (`l_log_parser` idx 10^2), and one is a genuine synergy break at `invoice_pricing` under one of its three draws ($I = +1.57$; the other two draws screen) — so screening at a point is a dominant-but-not-sure property of the alternative distribution, not a logical constant. Symmetrically, the three published mt6 break points *break again under every redraw* (9/9): breaks are properties of the decision point, not of the draw. Sixteen draws found no alternative (the selection channel of §7, rate consistent across schedules: 6/4/6). Among the screened points, 52 are non-trivially *shielded*: the harness flip alone has a real effect ($C(H) \neq 0$) that the joint flip erases — these are exactly the double-counting cases. Of the 52 shielded points, 24 carry constant-dictionary annotations that make the re-injection check possible (the remaining 28 are unannotated V2 points, which we declare rather than extrapolate). Among the annotated: **23/24** show *structural re-injection* — the sampled alternative action a' carries the destroyed information across the cut, a property of the estimand (since a' is sampled from the model conditioned on the un-summarized state); 9 also show the second mechanism and 1 shows neither. The second mechanism is checkable on all 52 shielded points: in **22/52** the *live downstream harness* re-triggers summarization anyway, equalizing the

²Point labels throughout are checkpoint indices (`cp_index`); raw artifacts also carry the 1-based tool-call `index` ($= \text{cp_index} + 1$), so this point appears as `index 11` in `results.jsonl`.

branches. Section 4.4 makes this division precise: a coalescence lemma and a one-line corollary guarantee that whenever the two branches coalesce, the resulting equality is exact — and we verify the occurrence directly in the recorded traces (the arguments explain 72/75 screened points; the 3 residuals, including 2 reward collisions, are declared). We also found genuine non-saturated synergy where the shield breaks (`1_vending_machine`: $C_H = C_M = -0.09$, $C_{HM} = 0$, so $I = +0.18$: either flip alone hurts, together they cancel).

Consequence for credit assignment. At every shielded point, single-layer harness credit $C(H)$ bills the harness for an effect the model absorbs: interaction-corrected credit $C_{HM} - C_M$ is zero while $C(H)$ is not. In the slack regime this is not an edge case — it is *all* measured points. Section 4.4 shows the error is not approximate: at screened points, $C(H) = -I(H, M)$ exactly.

4.4 Screening-off, formally

The measurements above are point identities, not expectations: under the premises of §4.1 (greedy decoding, fixed config, sequential serving, prefix caching off — with the resulting exactness *measured*, floor 0.0), the model, the environment, and the harness are all deterministic, so each replay branch is a single deterministic rollout. Fix a recorded trajectory with reward R_0 and a decision pair at index t : harness action h with alternative h' , model action a with sampled alternative a' . Writing $R_H = R(\text{do}(h'))$, $R_M = R(\text{do}(a'))$, $R_{HM} = R(\text{do}(h', a'))$ for the replayed rewards, the estimands are $C_H = R_0 - R_H$, $C_M = R_0 - R_M$, $C_{HM} = R_0 - R_{HM}$, and $I = C_{HM} - C_H - C_M$; throughout this section R is the test-fraction reward (not R_{eff} , whose token term accumulates over the differing prefixes), matching all measurements of §4.3. *Screening-off at the point* means $C_{HM} = C_M$.

Proposition 1 (Double-counting identity). *At any point, $C_{HM} - C_M = C_H + I$. Consequently, where screening-off holds: (i) the marginal credit of the harness flip in the presence of the model flip is zero, $C_{HM} - C_M = 0$; (ii) the single-layer harness credit equals the negated interaction, $C_H = -I$; (iii) a trainer that bills the harness C_H at such a point therefore assigns a signal consisting entirely of interaction — credit for an effect the model absorbs — while the interaction-corrected signal $C_{HM} - C_M$ vanishes.*

Proof. The first identity is the definition of I rearranged. Under $C_{HM} = C_M$, it gives $0 = C_H + I$; (i) and (iii) follow by definition. $\square$

Proposition 1 upgrades the empirical observation to a correction rule: the quantity a cross-layer trainer should bill the harness is $C_{HM} - C_M$ (one extra replay), which is exact in both regimes — equal to C_H where interaction is absent, and correctly zero where the model shields. Our training arm 3 uses precisely this signal.

Lemma 1 (Coalescence). *Let $s_k = (c_k, e_k)$ denote the agent state at step $k \geq t$ — model-visible context and environment (sandbox) state — and consider the two branches $B_M = \text{do}(a')$ and $B_{HM} = \text{do}(h', a')$. If the agent is deterministic and there exists $k \geq t$ with $s_k^{B_M} = s_k^{B_{HM}}$, then the branches coincide from k onward; in particular $R_M = R_{HM}$ and screening-off holds exactly.*

Proof. The one-step transition (model call, tool execution, harness policy) is a deterministic function of s_k . Equal states yield equal actions and equal successor states; induction to termination yields identical suffixes, hence identical final rewards. $\square$

Corollary 1 (Behavioral coalescence at the intervened turn). *If both branches execute identical action sequences from turn t to termination, then $R_M = R_{HM}$. (Harness context operations do not modify the environment, so the environment states at turn t agree; identical action suffixes applied to equal environment states under a deterministic environment yield equal final environment states, and R is a function of the final environment state.)*

The two empirical mechanisms of the anatomy are the two *coalescence-inducing events* we observe in practice; the lemma’s role is to guarantee that *whenever* coalescence occurs, the resulting equality is exact rather than approximate. Because reward equality does *not* certify coalescence (with reward resolution $1/n_{\text{tests}}$, collisions are possible), we verified the hypotheses directly in the recorded branch traces of all 75 screened points, comparing per-turn states (c_k, e_k) and per-turn behavior (action, observation), normalizing non-semantic identifiers that leak into test output (memory addresses, sandbox paths). The result, stated so that each count matches an enunciated argument: in **29/75** points the full state (c_k, e_k) becomes byte-identical within 1–4 turns of the intervention — Lemma 1’s hypothesis, verified; in **72/75** (a superset of the 29) the two branches execute identical action suffixes from the intervened turn itself — Corollary 1 applies and entails $R_M = R_{HM}$. The formal arguments therefore *explain 72 of the 75* measured screening events. The **3 residuals**: two (`l_log_parser` idx 10, in two configurations) are exactly the reward-collision case — $R_M = R_{HM}$ holds without trace coalescence, so the screening statistic counts them but the theory does not explain them — and one (`l_payroll_pipeline` idx 7 under g900, saturated at $R = 1.0$) coalesces behaviorally only 3 turns after the intervention, which is consistent with screening but not covered by Corollary 1; we declare all three rather than absorb them into the explained count. No suffix violated determinism: whenever states matched at some turn, the branches were identical from that turn to termination, corroborating the premises at trace level. *Re-triggering* (22/52): both branches executed the same a' , so environment states already agree; the live harness in B_M summarizes at step $t+1$, and context coalescence follows. *Re-injection* (23/24 annotated): a' (typically a `write_file` carrying the destroyed constants) equalizes the environment component, and the residual context difference is confined to content the greedy continuation never conditions on. The

lemma thus explains a striking feature of the data: measured screening-off consists of *exact zeros*, not small values — determinism converts coalescence into equality, the measured 0.0 floor makes equality observable, and the coalescence itself is now measured rather than presumed.

Remark 1 (Pivotality and the competence window). Call a decision point *reward-pivotal* if at least one of the three interventions changes the reward. Non-pivotal points satisfy $C_H = C_M = C_{HM} = I = 0$ identically: nothing is detectable there, by construction rather than by noise. When the agent fails a task outright and no intervention — single or joint — rescues it, every point on that trajectory is non-pivotal. This is the formal content of the competence window: reward saturation does not merely add variance — it removes pivotal points, which is why the Qwen3-1.7B pressure regime (11/12 saturated) had essentially no power and why the mt4 break rate collapses.

Corollary 2 (What single-layer training optimizes). *In a regime where screening-off holds at all measured points (our slack regime, 57/57), the per-decision signal of a single-layer harness trainer is $-I$ at every shielded point: it consists entirely of credit for effects the model absorbs. More generally, by Proposition 1 the single-layer signal errs by exactly I at every point where $I \neq 0$, regardless of regime — at the broken point `l_vending_machine`, $C_H = -0.09$ while the corrected signal is $+0.09$, an inverted sign. The corrected signal $C_{HM} - C_M$ is exact in both regimes.*

We emphasize the epistemic status of each piece. Proposition 1 is elementary algebra — its value is not mathematical depth but the correction rule it licenses. Lemma 1 holds given the determinism premises, and both the premises (§4.1) and the lemma’s hypothesis (this section) are *measured* rather than assumed. Whether coalescence occurs at a given point, which mechanism induces it, and how often screening-off holds at all are empirical questions that the rest of this section answered.

4.5 Control tasks and the `c_temp_label` exception

On control tasks, where summaries destroy nothing by construction, the method returns $C(H) = 0$ in 11/12 measured points — the construct behaves. The exception is informative: `c_temp_label` showed $C = +0.22$ at turn 0 despite all task-critical information surviving the summary. Audit of the replay pair shows the effect is *behavioral*, not informational: the summarized context changes the model’s subsequent formatting behavior, which a strict test then penalizes. $C(H)$ measures total causal effect through *any* pathway; the controls certify the informational pathway is clean, and the exception demonstrates the behavioral pathway exists and is detectable.

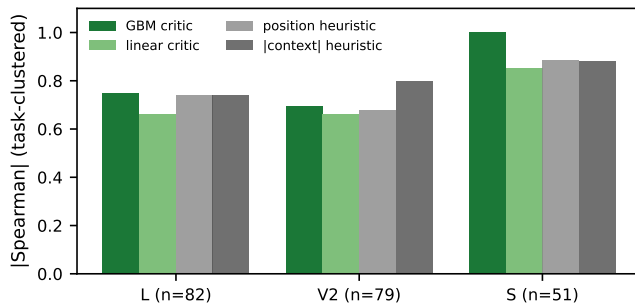


Figure 4: **F4 — Critics vs. trivial heuristics per stratum** (task-clustered $|\text{Spearman}|$). The critic never beats the best heuristic; $C(H)$ is structured by position and context size.

5 The Structure of Harness Credit

Can $C(H)$ be *predicted* from cheap features, so that replay is needed only to calibrate? We trained linear and GBM critics on 248 ground-truth points (features: turn, context size, position, test progress; task-clustered evaluation) against the baselines mandated by [14]: position and context-size heuristics. The honest answer is that **the critic never beats the heuristics** (Figure 4): on the combined L+V2 strata ($n = 161$, task-clustered Spearman) the GBM reaches 0.667 and the linear critic 0.675, against 0.708 for the position heuristic and 0.770 for $|\text{context}|$; pooled over all 248 points the linear critic’s 0.718 still trails the context-size heuristic. Per stratum, L is a tie (GBM 0.747 vs. position 0.741), in V2 the heuristic wins (0.693 vs. 0.796), and S saturates for everything (AUROC 1.0). For detection ($|C| > 0$) the GBM has a modest edge (AUROC 0.846 vs. 0.735 for position and 0.785 for the sign-inverted context-size heuristic) with overlapping CIs.

We report this not as a failed component but as a finding about the signal: **harness credit is mostly a function of context size and position**. That is exactly what the information-destruction mechanism predicts — how much is destroyed, and how late — and it means practitioners can get most of the ranking value of $C(H)$ from features they already log, reserving replay for calibration and for the interaction term, which no cheap feature predicts. The cross-environment test sharpens this: a critic trained on all $C(M)$ ground truth from the designed tasks and evaluated on the 94 MBPP+ $C(M)$ points (the only quantity with data on both sides; $C(H)$ has none in the MBPP+ runs) transfers at chance (Spearman 0.03 linear / -0.08 GBM, AUROC 0.30/0.50, MAE worse than a constant predictor; pre-registered secondary of D3). Amortized credit does not cross environments here — consistent with the replay-ground-truth bar of [14], and a further reason the training signal should be anchored in replay rather than in a learned proxy.

6 Training with Cross-Layer Credit

Does better credit train a better harness? We train the harness `context_policy` (a logistic policy over 5 features: bias, turn fraction, context tokens, test progress, message count) with four arms, **dose-matched in total rollouts** (episodes + credit replays): (1) *outcome-only* policy gradient; (2) *single-layer* $C(H)$ credit — an explicit reproduction of the CHILL-Harness principle in our environment; (3) *interaction-corrected* credit $C_{HM} - C_M$; (4) a *zero-credit* masking control (credit $\equiv 0$, no replays — which under our greedy tie-break at $\theta = 0$ is de facto an untrained keep-always policy; we state this explicitly). The pre-registered success criterion (GATE 3) required arm 3 to beat both arm 2 and arm 4. Calibration established the target is learnable in principle: on held-out tasks a simple context-size threshold policy (thr600) achieves $R_{\text{eff}} = 0.398$, dominating keep-always (0.237) and summarize-always (−0.464); thr600 is expressible by the policy class.

Act 1: seed-dependent collapse. With raw features and $\text{lr} = 0.5$, every trained arm collapsed into one of the two fixed policies, with the attractor determined by seed (outcome: k/k/s; $C(H)$: s/s/k; $C_{HM} - C_M$: k/s/s). GATE 3 failed and we report it. Two hypotheses remained: the credit estimator is broken, or the optimization is. A pre-registered audit recomputed 60 logged credits by fresh two-greedy-replay: **58/60 sign agreement** (mean discrepancy −0.0015). Defining a *harmful flip* as logged-positive/recomputed-negative — a definition we state here, the registration specified sign agreement only — there are zero: the two disagreements are one benign flip (`inventory_restock` idx 0, −0.084 $\rightarrow$ +0.10; logged as harmful, actually helpful) and one zero-to-negligible boundary case (`c_temp_label` idx 6, 0.0 $\rightarrow$ −0.001). The estimator is exonerated.

Act 2: stable optimization, no discovery. With the pre-registered fix (feature centering, $\text{lr} = 0.1$, gradient clipping), training is sane: gradients bounded, learned weights point the right way (negative bias, positive weight on context tokens, implicit decision thresholds at 2.5k–5.8k tokens). Yet **no arm in any seed discovers the threshold policy**: all non-collapsed runs converge to keep-always (held-out $R_{\text{eff}} = 0.237$ vs. thr600’s 0.398; Figure 5). Credit-driven arms also retain collapse risk the outcome-only arm avoids ($C(H)$: 1/3 seeds; $C_{HM} - C_M$: 1/3; outcome-only and zero: 0/3) — a seed-count difference that is anecdotal at $n = 3$ (Fisher $p = 1.0$) and that we report without testing.

Diagnosis. Dose-matching in total rollouts means credit arms pay for replays with episodes: across 3 seeds the outcome arm saw 1,789 training episodes, $C(H)$ 786, $C_{HM} - C_M$ 319 (zero: 1,910). One might suspect the credit arms simply saw too little data. The state-visitation and reward-margin records rule this reading out and localize the failure in three steps. *First*, the learned poli-

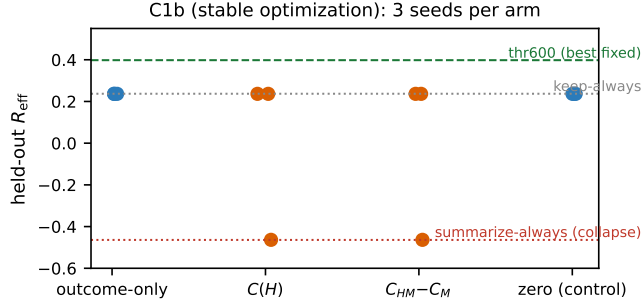


Figure 5: **F5 — Training outcome (C1b, stable optimization).** Held-out R_{eff} , 3 seeds per arm. No arm reaches the expressible thr600 policy; credit-driven arms (orange) additionally collapsed in 1/3 seeds each (anecdotal at $n = 3$).

cies are behaviorally inert: their implicit decision thresholds sit at 2,500–5,800 context tokens, but no decision in any arm or seed — 0 of 13,498, including the zero arm, de facto keep-always, the policy that maximizes context growth (1,910 episodes, cap 1,532 tokens) — and no held-out decision (cap 1,200 tokens) ever reaches that region: the learned thresholds can never fire, so every non-collapsed policy is keep-always in behavior. *Second*, in the states the training tasks *do* visit (thr600 fires above 600 tokens, well inside the visited range), the reward margin between thr600 and keep-always is statistically indistinguishable from zero: paired bootstrap on the training tasks gives +0.024, CI95 $[-0.029, +0.079]$, $P(\text{diff} \leq 0) = 0.19$ (re-derivable from the raw calibration artifact by the bundled `margem_calibracao.py`, which fails on drift) — whatever gradient signal exists on the training tasks is bounded by this margin, which is undetectable at this sample size; we note the interval admits margins (up to +0.08) that more episodes might resolve, so this second argument is an underpowered null and the diagnosis rests primarily on the state-visitation argument. *Third*, thr600’s held-out dominance comes from cumulative-cost structure: long held-out tasks accumulate prompt tokens under keep-always (e.g. `l_vending_machine`, 5,933 cumulative prompt tokens, $R_{\text{eff}} = -1.12$) — a cost regime the training distribution never makes consequential. The train/held-out split was alphabetical, fixed before training and before we knew this gap existed. **The binding constraint is the training task distribution, not credit quality.** This sharpens the design lesson: a cross-layer training benchmark must verify that its training tasks assign a nonzero reward margin to the states where candidate policies actually differ — otherwise credit fidelity is moot, and the failure will masquerade as an algorithmic one.

7 Threats to Validity and Discussion

7.1 External validity

Our tasks are designed, not sampled from a public benchmark, and this is a real limitation — but a motivated one. The measurement requires (i) control over the *position* of destroyable information, to separate informational from behavioral effects; (ii) per-stratum calibrated recoverability, which underlies the control tasks; and (iii) deterministic fractional reward, which underlies the exact floor. No public benchmark provides these jointly. We therefore ran the missing experiment in reduced form (pre-registered as D3 before running): a multi-turn MBPP+ adaptation — 60 tasks derived mechanically from MBPP+ [8] with fractional reward (6–10 test functions per task, canonical solutions validated in our sandbox), one slack and one pressure configuration on Qwen3-4B. The floor replicates exactly (1,294 null replays, all exact) and **screening-off replicates at 66/66 interaction points, of which 11 are reward-pivotal (11/11 screened)** — the other 55 screen by construction (Remark 1), so the informative content of the replication is the pivotal 11/11, not the 66/66 — including a non-saturated point with $I = -0.4$, a live double-counting case in an external benchmark. What the adaptation cannot yet test is the pressure-regime break rate: 60/66 points are reward-saturated (Qwen3-4B is too strong for MBPP+), leaving 6 non-saturated points and no breaks — inconclusive at that power, the same saturation boundary seen at 1.7B and 8B. A pre-registered second attempt (D6: the 1.7B on the same MBPP+ pool, declared outcomes) landed on the same boundary from the same side: floor exact (1,080/1,080), screening replicates raw at 44/44 points across both regimes (24/24 reward-pivotal) — a second (model, environment) cell with unconditional screening in the external benchmark — but 20/22 points per configuration are saturated (the 1.7B still solves 34/60 MBPP+ tasks at 1.0) and the pressure budget never binds on MBPP+’s short trajectories, so the pressure cell of environment B remains empty after two attempts, and we declare it rather than stretch it. The designed tasks remain necessary for the regime-dependence claim; the core phenomenon — exact floor, exact screening, double counting — is not an artifact of our task design. We position this as a measurement paper: the transferable artifacts are the recorder/replay/forced-queue infrastructure, the APC finding, and the measurement protocol.

7.2 Replication across models, pools, and environments

Floor and unconditional replications. The floor replicates everywhere (pre-registered as D2c, D2d, D3 before running): the 8B contributes 810 dedicated and 234 attached null replays and MBPP+ contributes 1,294, all exact. Screening-off replicates *unconditionally* on Qwen3-1.7B (0 raw breaks in 23 points) and on MBPP+ (66/66 raw; 11/11 pivotal).

The 8B breaks the shield. On Qwen3-8B it does not: raw break rates are flat across regimes (5/28 slack, 6/34 mt6, 7/35 mt4), so the slack/pressure contrast established on Qwen3-4B does *not* transfer to the 8B, and we report this as the headline of the replication rather than conditioning it away. Saturation-based exclusion would be invalid here: clipping R at $\{0, 1\}$ can only *shrink* $|C_{HM} - C_M|$, so saturation can mask nulls but cannot fabricate breaks — a saturated break of 0.9 is a real break. The 8B slack breaks have a sharp common structure ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \in [0.8, 0.9]$, hence $I \in [-0.2, -0.1]$): the harness flip alone is catastrophic, the model flip alone is null, and the joint flip tracks the harness — the shield *inverts*. The mechanism is the one the anatomy of §4.3 identified: on the same tasks the 4B’s sampled a' re-injects all destroyed critical constants (5/5, 5/5, 6/6 on the shielded points), while the 8B’s a' re-injects none (0/5, 0/6, 0/8, 0/5; matching by constant-literal substring in the serialized a') — its solutions are structured so that the alternative action no longer carries the constants. The mechanism is verifiable on 4 of the 5 slack breaks (the fifth, `invoice_pricing`, has no annotated constants), and the breaks *recur*: all 5 slack break tasks break again, with identical per-task structure, in both pressure configurations (6/34 at mt6, 7/35 at mt4). Whether screening-off holds is therefore itself competence-dependent: at that stage of the design the shield appeared to be a property of the model–harness pair — a claim the symmetric control below narrows further to the (model, harness, task) triple (the closest precedent, model-dependent counterfactual *measurability* [14], concerns whether ground truth can be estimated at all; here the recovered causal structure itself inverts). We say competence, not scale: 4B and 8B differ in post-training as well as size, and the claim is anchored in the measured re-injection contrast on the original pool (which, as reported below, does not generalize as a mechanism), not in parameter count. Under pressure the 8B also shows a genuine non-saturated synergy break at `1_door_controller` ($C_{HM} = 0.23$ vs. $C_M = -0.46$, $I = +0.38$). Saturation still dominates both auxiliary models in opposite directions (11/12 of the 1.7B’s pressure points, from weakness; 90/97 of the 8B’s points, from strength — and the slack regime leaves the 8B a single non-saturated point, so the conditioned slack contrast has essentially no power and we do not claim it).

Curated pool (pre-registrations 17–18). The remaining step — the break rate on a pool curated to the 8B’s competence — was then run (pre-registrations 17 and 18): a first hardened pool failed its pre-registered curation rule (17/24 tasks solved at 1.0; reported as failure), and a second pool targeted at the fractional-reward profile passed (22 of 44 tasks in the (0.05, 0.95) window; 792/792 nulls exact after the ASLR fix of §4.1). **In-window, the 8B’s raw screening breaks replicate and stay flat across regimes** — 3/27 slack, 3/31 pressure — including a *non-saturated slack break* (`x_hours_bank`: $C_H = +0.25$, $C_M = -0.17$, $C_{HM} = 0.00$, $I = -0.08$), which closes the clipping objection: breaks persist where saturation cannot manufacture them. The pool-matched competence contrast is D2d’s (same 30 tasks: 4B 0/57 raw vs. 8B 5/28); within

the curated pool the regime contrast seen on the 4B does not exist on the 8B even inside its competence window. Two of the three break points recur across both regimes with identical structure. Two pre-registered sub-hypotheses failed and we report them as such: the binary re-injection mechanism does not generalize (at active slack points the curated-pool *a'* *never* re-injects constants, 0/15, yet only 3/15 break; under pressure 2 of 3 breaks occur *with* re-injection; one-sided Fisher $p = 1.0$ and $p = 0.94$ — the slack table is degenerate, no exposed points, so only the pressure test is informative, and it points the wrong way), and the directional prediction (pressure $\geq$ slack) failed (9.7% vs. 11.1%). All curated-pool break counts are descriptive: with 3 break tasks per regime we make no clustered inferential claim, and we make no cross-pool p -value comparison against the 4B’s 0/57.

Symmetric control (pre-registration 19) and the triple claim. Two design threats were named at this point — the curated comparison was cross-pool (the 4B’s 0/57 comes from the original pool — the paired contrast on identical tasks is D2d’s 0/57 vs. 5/28); and the curation window is operationalized by task *family* (the second pool was designed around the profile that survived curation), so in-window break rate could be a property of the family rather than of competence — and the pre-registered symmetric control (D5: the 4B on the same 22 curated tasks, outcomes declared in advance) resolved both, *against* the strong form of our own claim. The 4B also breaks raw screening on the curated pool: 4/16 slack points (3 non-saturated, including a synergy break at `h_turnstile_fsm` with $I = +0.77$ and `h_hotel_folio` at $I = -1.42$) and 2/23 pressure points; its floor held (396/396 nulls exact), and the pivotality-based prediction that the 4B would sit saturated-in-failure on this pool failed (1/16 and 3/23 saturated — reported as a failed pre-registered sub-hypothesis). The competence claim therefore re-scopes: the shield is a property of the (model, harness, *task*) triple, not of the model–harness pair alone.

What survives every cell. The pool-matched model contrast on identical tasks (0/57 vs. 5/28), the majority of points screened in every configuration (worst cell 12/16), the regime contrast existing only for the 4B on its original pool, and a majority pattern of break-point stability: break sets are disjoint across models at the point level (the two models share one break task, `h_hotel_folio`, at different turns), and within a model most break points recur across regimes — in the 8B, 2 of 3 slack break tasks recur numerically identically; in the 4B, 2 of 3 break tasks recur, one numerically identically (`h_hotel_folio`), one with the same sign structure but different magnitudes (`h_turnstile_fsm`: $I = +0.77$ slack vs. $+0.31$ pressure), and `h_telco_roaming` does not recur. A further reading — breaks concentrate where the pool sits near the model’s competence boundary (for the 4B, the original pool under pressure and the hardened pool in both regimes; for the 8B, the hardened pool in both regimes) — is descriptive and not pre-registered: the pre-declared interpretation of outcome c2 was “a property of the task family, not of competence,” and what favors

the triple reading over the pure-family reading is the point-level disjunction of break sets between models on the identical pool. The break structure is also heterogeneous — the D2d slack signature ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \approx 0.9$) does not recur — and a descriptive trace anatomy of all 8B and 4B curated-pool breaks (0 new rollouts) finds one uniform base, three recurring patterns, and a model asymmetry (restore-direction flips appear only in the 4B); Appendix C reports it in full, including two audit details showing that recurring break points are numerically identical across regimes while *I magnitudes* are baseline-relative. The robust claim is therefore narrower than the mechanism story: *whether* screening-off holds is a property of the (model, harness, task) triple, and re-injection of destroyed information, which fully explained the annotated 4B/8B contrast of D2d, is neither necessary nor sufficient for the shield in a curated pool.

7.3 Statistical caveats

Clustered inference. Points within a task are not independent. At the task level the pressure-vs-slack contrast is 3 tasks vs. 0 (sign test $p = 0.125$); at the point level $p = 0.005$; the saturation-conditioned contrast behaves the same way (point-level $p = 0.037$; task-clustered permutation $p = 0.18$ – 0.25). We report all levels and lean on the pivotal-point screening-off sign test (53/56, $P = 4.1 \times 10^{-13}$; 19/22 after collapsing to unique points) and cross-configuration recurrence of break points as the stronger evidence.

Multiplicity. This paper reports on the order of a dozen p -values across regimes, strata, and clustering levels, and we apply no family-wise correction. Only the pre-registered outcomes (Table 4) are confirmatory; borderline point-level values ($p = 0.037$, $p = 0.005$) would not all survive a family correction, and we treat them as descriptive support. The claims we retain rest on replicated structure — exact screening at pivotal points, cross-configuration recurrence of break points, and the pool-matched model contrast — rather than on any single threshold crossing.

Saturation is endogenous, and conditioning on it is post-treatment. Reward resolution is set by test count, so saturation is partly a design artifact. Worse for inference: saturation is defined on the rewards of the replayed arms, i.e. on outcomes of the intervention itself, so every saturation-conditioned rate in this paper conditions on a post-treatment variable and carries selection bias of unknown sign. We therefore use conditioned contrasts descriptively only, never confirmatorily, and we describe saturation as a correlate of the competence window, not a mediator. The inverted-U at mt4 is only partially accounted for (conditioned rate $0.045 \rightarrow 0.143$; the ordering $\text{mt8} > \text{mt6} > \text{mt4}$ persists), so the window remains a partially supported hypothesis, discussed but not claimed as a contribution.

7.4 Estimand and infrastructure caveats

Estimator coupling. $C(M)$ and I share the sampled alternative action a' : they are not independent evidence, and the structural component of screening-off (§4.3) is partly a property of the estimand — a' sampled from the unsummarized state can re-inject information. We quantified this (23/24 annotated shielded points; 28 unannotated points declared) rather than hiding it; an estimand with a' sampled from the summarized state would measure a different, also legitimate, quantity. Two further limitations of the estimand: a' is a *single* sample per point (fixed seed schedule, §3) in the census; the pre-registered re-sampling study (§4.3) measures sensitivity to that draw directly and finds the slack result stable (99/101) and the pressure breaks draw-invariant (9/9). What remains unmeasured is coverage of the alternative *distribution* beyond $k = 3$ redraws at temperature 0.8, and the alternative estimand with a' conditioned on the summarized state.

Selection into measurement. A point enters the interaction census only if the sampler finds a parseable $a' \neq a$ within the 8-seed schedule; across configurations 53–73% of candidate points are discarded this way. The measured population is therefore selected on the model’s own output entropy at that state — a post-decision variable plausibly correlated with both competence and the information content of the context — and every break or screening rate is conditional on “an alternative exists.” For the headline cross-model contrasts the discard rates are close enough that differential selection is an implausible driver: on the pool-matched contrast (identical 30 tasks) the 4B discards 66–73% across its slack configurations versus 64% for the 8B at the same budget, a ≤ 9 -point selection gap against a 0% vs. 18% raw break-rate gap; on the curated pool the rates are 57–70% (4B) versus 53–59% (8B). We measured the composition directly (`analise_selecao.py`, from the per-point sampling logs joined to the baseline trajectories): selection is real and directional — in *both* models the retained points sit earlier in the trajectory (pool-matched relative position, found vs. discarded: 0.33 vs. 0.73 for the 4B, 0.21 vs. 0.79 for the 8B; task-clustered bootstrap CIs exclude zero) and carry smaller contexts. But the direction *disfavors* the headline contrast rather than explaining it: the 8B’s retained points are shallower than the 4B’s (mean turn 0.65 vs. 1.55, context 395 vs. 433 tokens, $d = 0.31$ – 0.54), so if depth promotes breaks, composition biases the 8B toward *fewer* measured breaks — and it still breaks at 18% against the 4B’s 0%. What selection does forbid is any claim about deep-context interaction: no census measures points the sampler cannot reach, and the discarded majority is precisely the late-trajectory, large-context population.

Serving premises. The exact floor holds under a fully specified stack: vLLM 0.8.5.post1 (V1 engine, pinned; PyTorch 2.6.0+cu124, CUDA 12.4, driver 580.173.02, Python 3.11.14, full transitive closure pinned in the repository’s `uv.lock`) serving one model per experiment on a single NVIDIA RTX 4090; dtype auto (bfloat16); CUDA graphs enabled (no `enforce_eager`); default chunked pre-

fill (`max_num_batched_tokens` = 2048); automatic prefix caching explicitly disabled; GPU memory utilization 0.85 (0.92 for the 8B); strictly sequential requests against a single instance; greedy decoding (temperature 0, single seed) for every episode and replay rollout; and the fixed a' seed schedule of §3. The floor is a measured property of this stack, not a theorem; any deviation (batching, caching, config drift, a version bump) voids it and must be re-measured.

Protocol cost. Exactness is bought with throughput: the premises force strictly sequential serving, one request at a time, on one GPU. The full measurement inventory is 7,685 null replays, 1,380 counterfactual replays at 460 interaction points (three per point), 294 single-flip $C(H)$ replays, 304 re-sampling replays with 401 a' sampling calls (pre-registration 21), and 1,617 census a' sampling calls (up to 8 seeds each); the training comparison adds 4,804 episodes (Act 2) plus the dose-matched replays. The chained experiment logs account for roughly 25 wall-clock hours on the single RTX 4090 (an under-count: the earliest runs predate chained logging). Dose-matching makes the replay tax concrete: at equal rollout budget, the interaction-corrected arm afforded 319 training episodes against the outcome-only arm’s 1,789 — a $5.6\times$ episode cost that any replay-based trainer must amortize.

Beyond determinism: the stochastic protocol. Real agent stacks violate every exactness premise (sampling temperature, flaky tests, batched serving). The protocol degrades gracefully rather than breaking: under stochastic replay each branch reward becomes a distribution, $C(d)$ a difference of means, and the measured null-replay *distribution* replaces the exact floor. The per-point recipe is: (i) estimate the null spread from n_0 null replays (the analogue of §4.1, now a variance rather than a maximum); (ii) size the per-branch replay count so that the minimal credit of interest exceeds the resulting standard error; (iii) cluster inference by task, as here. Exact screening-off becomes a test of $C_{HM} - C_M$ against the null band rather than a literal equality, and the coalescence lemma weakens to a coupling argument; what survives unchanged is the double-counting identity, which is algebraic and holds in expectation. The deterministic setting of this paper is the calibration case where all of this is measurable at zero variance — we deliberately paid throughput for that identifiability.

What the triple claim does to the practical prescription. If *whether* screening holds depends on the (model, harness, task) triple, no fixed correction transports across stacks: “measure before training” survives only as a per-stack diagnostic — run the null floor, then the screening census on your own triple, then decide which credit signal is billable. This deliberately weakens the prescription from a rule to a protocol. The transportable artifacts are the identity $C(H) = -I$ at screened points (algebra, stack-independent), the correction rule $C_{HM} - C_M$ (one extra replay), and the census procedure itself — not any particular census result.

8 Conclusion

We built the first apparatus that measures, on the same trajectory, the causal credit of harness decisions, model decisions, and their interaction, against an exactly-zero measured noise floor. The measurement yields a crisp, regime-dependent picture: with budget slack the model screens off the harness completely (0/57, stable under pre-registered re-sampling of the alternative action), so single-layer harness credit double-counts; under budget pressure the shield breaks at recurring points (8/65), inside a competence window shaped by reward saturation — and the regime contrast itself is a property of the (model, harness, task) triple: a stronger model breaks screening at a flat rate in both regimes, and on a pool hardened to the stronger model’s competence boundary *both* models break, at model-specific, largely regime-stable points. Better credit did not, however, train a better harness: a two-stage, pre-registered negative — with the estimator exonerated by audit — locates the binding constraint in the training task distribution itself: no reward margin where policies differ, unreachable states where learned thresholds would fire. We offer the infrastructure, the regime-dependence finding, the APC determinism warning, and a pre-registration ledger of 21 predictions (including 4 failures and 6 partials) as the contributions, and the training negative as the design brief for the benchmark this field still needs.

A Pre-Registration Ledger

All confirmatory analyses were pre-registered in the project’s execution plan before the corresponding data were collected, with one dated exception. The ledger is an *internal, same-team* pre-specification, timestamped by git commits; for every item but one the registration commit precedes the corresponding run artifacts. The exception is item 10: the first slack census (g600) completed 34 minutes *before* the registration commit, so we treat g600 as discovery data and report the discovery/confirmation split in §4.3 (confirmation alone: 39/42). It is not a third-party registration, and minutes-to-hours gaps in a single-team pipeline cannot exclude unreported exploration before each commit (the pilot runs that preceded the formal ledger are preserved in the artifact tree). Its evidential force rests on the recorded failures, not on the registration mechanism itself. Table 4 lists all 21 pre-registrations with their outcomes, including the four that failed and the six partial. Failed predictions are reported in the main text where they occur.

B Null-Replay Reconciliation

Table 5 reconciles the canonical null-replay total (7,685, all exact) against every run artifact; a bundled script re-derives the table and fails if any count drifts. Excluded runs are listed with their reasons: the ASLR pre-fix runs (premises

held, fidelity broken by an unsanitized memory address; root-caused and re-run) and the APC-contaminated run (premises violated by design, kept as the Figure 1b demonstration).

C Break-Point Anatomy (Descriptive)

This appendix reports the full descriptive trace anatomy of the curated-pool screening breaks (0 new rollouts; all from recorded replay traces). It is descriptive throughout: no count below is a pre-registered test.

The 8B’s six breaks. One uniform base and two patterns on top of it. Uniformly: every break is a `keep`→`summarize` flip at a `write_file` turn, the summary destroys essentially all critical constants in context (8/8, 5/6, 7/7, 10/10), and the harness flip alone is destructive. On top: either the sampled a' carries some constants and partially rescues the destroyed context (the joint flip lands *between* the single flips: `h_hotel_folio`, `h_cargo_manifest` — the former tracks the model, $C_{HM} = 0.08$ vs. $C_M = 0.00$; the latter the harness, $C_{HM} = 0.46$ vs. $C_H = 0.62$), or a' is itself beneficial and the destruction annuls exactly that benefit (the joint flip returns to the original reward: `x_hours_bank`, `h_sku_validator` — the one positive I is of this kind). The two recurring points are numerically identical across regimes — same four arm rewards, same destroyed constants, same free-divergence offsets, from different baseline trajectories — so the break is a property of the decision point (task $\times$ turn), not of the budget regime.

The 4B’s D5 breaks. The same anatomy adds a structural asymmetry: two of the 4B’s four unique break points (three of the six break instances, counting the recurring `h_turnstile_fsm` point) are *restore*-direction flips (`summarize`→`keep`: the baseline had already summarized, and the flip restores destroyed constants), a direction absent from the 8B’s breaks, and they follow a third pattern — individually beneficial or neutral flips whose joint application destroys the benefit ($I > 0$). Only one 4B point falls into an 8B pattern.

Audit details. The recurring `h_hotel_folio` point is numerically identical across regimes in all four arm rewards despite distinct baseline trajectories, and the recurring `h_turnstile_fsm` magnitude difference ($I = +0.77$ slack vs. $+0.31$ pressure) traces entirely to baseline quality — C_{HM} and all free-divergence offsets are identical across regimes. The I *magnitude* at a recurring point is therefore baseline-relative even when the break itself is stable.

References

- [1] Yanjun Chen, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, and Wei Zhang. Exact is easier: Credit assignment for cooperative LLM agents. arXiv:2603.06859v2, 2026.
- [2] Zhengyu Chen, Teng Xiao, Huaisheng Zhu, Yige Yuan, Luan Zhang, and Jingang Wang. Co-Harness: Co-evolving harnesses and model weights for LLM agents. arXiv:2607.22688, 2026.
- [3] Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.
- [4] Jiarun Fu, Lizhong Ding, Sida Chen, Honglei Xin, Chunhui Zhang, Pengqi Li, Qiuning Wei, Ye Yuan, and Guoren Wang. CHILL-Harness: Counterfactual harness learning for efficient reasoning in long-horizon agents. arXiv:2607.25825, 2026.
- [5] Anna Harutyunyan, Will Dabney, Thomas Mesnard, Mohammad Gheshlaghi Azar, Bilal Piot, Nicolas Heess, Hado van Hasselt, Greg Wayne, Satinder Singh, Doina Precup, and Rémi Munos. Hindsight credit assignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [6] Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference. Thinking Machines Lab: Connectionism blog, 2025.
- [7] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- [8] Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by chatGPT really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [9] Haochen Luo, Yi Huang, Sichun Luo, Fengyuan Liu, Lei Li, Zefa Hu, Junlan Feng, and Qi Liu. Harness-aware self-evolving: Co-evolving model weights, harness, and task solutions. arXiv:2607.03935, 2026.
- [10] Xufang Luo et al. Agent lightning: Train ANY AI agents with reinforcement learning. arXiv:2508.03680, 2025.
- [11] Hans Reichenbach. *The Direction of Time*. University of California Press, 1956.
- [12] Jaineet Shah. Causal agent replay: Counterfactual attribution for LLM-agent failures. arXiv:2606.08275, 2026.

- [13] David H. Wolpert and Kagan Tumer. Optimal payoff functions for members of collectives. *Advances in Complex Systems*, 4(2–3):265–279, 2001.
- [14] Haiyue Zhang. Credit without ground truth: Auditing step-level credit assignment in LLM agents against executed replay. arXiv:2608.19760, 2026.
- [15] Luan Zhang, Ruochen Zhou, Dandan Song, Zhengyu Chen, Yuhang Tian, Jun Yang, Huipeng Ma, Chenhao Li, Guangyuan Feng, et al. Harness-Compass: Guiding automatic harness evolution toward generalizable and effective agent harnesses. arXiv:2608.01918, 2026.

Claim	Evidence	Status
Replay noise floor is exactly zero	7,685/7,685 nulls exact, 3 models $\times$ 2 envs (Table 5)	confirmed (pre-reg 1)
vLLM APC breaks greedy determinism	floor 0.417, 7/270 inexact	demonstrated
Slack screening-off, 4B original pool	0/57 raw; pivotal 53/56 = 0.946, CI95 [0.87, 1.00] task-clustered	confirmed (pre-reg 10, re-specified)
Screening survives a' re-sampling	99/101 informative slack draws (CI95 [0.94, 1.00]); mt6 breaks persist 9/9	confirmed (pre-reg 21)
Screening survives discovery/confirmation split	discovery (g600) 14/14; confirmation 39/42; strictly post-registration 27/30	confirmed (split forced by ledger audit)
Regime contrast not explained by selection composition	retained 8B points <i>shallower</i> than 4B's (turn 0.65 vs. 1.55); direction disfavors the contrast	supported (§7)
$C(H) = -I$ at screened points	algebra (Prop. 1); exact in all measured slack points	identity
Pressure breaks recur at fixed points (4B)	8/65; same points across configs	point-level; task-clustered weak ($p = 0.125$)
Break rate monotonic in pressure	inverted U, Cochran–Armitage $p = 0.38$	failed (pre-reg 13)
Saturation mediates the window	conditioned rates only	descriptive (post-treatment)
Screening-off is universal across models	1.7B: 0 raw breaks; 8B: breaks flat in both regimes	refuted
Shield is a property of the (model, harness, task) triple	curation + symmetric control; break sets point-level disjoint across models	supported; positive characterization inductive
Re-injection mechanism generalizes	23/24 annotated (original pool) vs. 0/15 exposed (curated), Fisher $p = 1.0/0.94$	failed (pre-reg 18)
Screening replicates in a second environment	MBPP+: 11/11 pivotal (4B), 24/24 pivotal (1.7B); rest screens by construction; pressure cell empty	confirmed; pressure inconclusive (pre-reg 16, 20)
Interaction-corrected credit trains a better harness	no arm learns thr600; margin +0.024, CI95 [−0.029, +0.079]	failed (pre-reg 9, 12); cause identified
Learned critic beats cheap heuristics	context-size heuristic wins overall	refuted; $C(H) \approx$ position + size
Amortized credit transfers across environments	Spearman 0.03/−0.08, chance	refuted (pre-reg 16)

Table 1: T0 — master claim ledger: every claim, its evidence, and its epistemic status. Bold **failed/refuted** rows are reported as results, not omissions. A bundled **make reproduce** re-derives the null, census, split, dose-response, margin, and selection statistics from the raw artifacts and fails on drift.

Method	counterfactual via replay	measures $C(H)$	measures $C(M)$	$I(H, M)$ on same trajectory	trains with I	structure across models tested
CHILL-Harness	✓ ^{off}	✓	—	—	—	—
CAR	✓	—	✓	— ^s	—	—
C3 v2	✓	(per agent)	—	—	—	—
Replay audit [14]	✓	—	✓	—	—	~ ^m
Co-Harness / HASE	—	—	—	—	— [*]	—
Ours	✓	✓	✓	✓	tested [†]	✓

Table 2: T1 — positioning. ^{off}Checkpointed paired replay used offline to train an amortized estimator; disabled at decision time. ^sCAR measures step-step interaction (Shapley) *within* one layer, not across the model/harness boundary. ^{*}Joint harness+weights training without per-decision causal measurement. ^mObserves that counterfactual *measurability* is model-dependent, within a single layer; no cross-layer structure. [†]Tested with dose-matched controls and pre-registration; the outcome is an identified negative (§6).

Model	Pool	Slack	Pressure	Reading
4B	original (30)	0/57	8/65	breaks gated by pressure
1.7B	original (30)	0/23	combined	slack replicates; pressure saturated
8B	original (30)	5/28	6/34, 7/35	flat: regime-independent breaks
4B	curated (22)	4/16	2/23	breaks in both regimes
8B	curated (22)	3/27	3/31	flat again
4B	MBPP+ (60)	0/66	(11/11 pivotal)	screens unconditionally
1.7B	MBPP+ (60)	0/44	(24/24 pivotal)	screens unconditionally

Table 3: **Synthesis: raw screening-off break counts by (model, pool, regime).** Every claim about *when* the shield holds reduces to this grid: pressure-gating is a 4B-on-original-pool phenomenon; the 8B breaks at a flat rate wherever it is in its competence window; MBPP+ screens everywhere. Counts are point-level (breaks/points); per-configuration detail in Figure 3.

#	Pre-registration prediction / rule	Outcome	Status
1	None. Break per config is none (all 0 over null).	0.0 to all 23 declared + 21 attached over (7.66) null. (table 1)	held
2	Pressure A holds / competitive win.	observed as expected.	held
3	Pressure A holds / competitive win.	observed as expected.	held
4	Pressure A holds / competitive win.	observed as expected.	held
5	Pressure A holds / competitive win.	observed as expected.	held
6	Pressure A holds / competitive win.	observed as expected.	held
7	Pressure A holds / competitive win.	observed as expected.	held
8	Pressure A holds / competitive win.	observed as expected.	held
9	Pressure A holds / competitive win.	observed as expected.	held
10	Pressure A holds / competitive win.	observed as expected.	held
11	Pressure A holds / competitive win.	observed as expected.	held
12	Pressure A holds / competitive win.	observed as expected.	held
13	Pressure A holds / competitive win.	observed as expected.	held
14	Pressure A holds / competitive win.	observed as expected.	held
15	Pressure A holds / competitive win.	observed as expected.	held
16	Pressure A holds / competitive win.	observed as expected.	held
17	Pressure A holds / competitive win.	observed as expected.	held
18	Pressure A holds / competitive win.	observed as expected.	held
19	Pressure A holds / competitive win.	observed as expected.	held
20	Pressure A holds / competitive win.	observed as expected.	held
21	Pressure A holds / competitive win.	observed as expected.	held
22	Pressure A holds / competitive win.	observed as expected.	held
23	Pressure A holds / competitive win.	observed as expected.	held

Table 4: Pre-registration ledger. Failures are results, not omissions: #9 and #12 constitute the two-stage training negative (§6); #13 led to the competence-window analysis (§4.3).

Run	Model / pool	Dedicated	Attached	Inexact
g450, g600, g900	4B / original, slack	3×270	64, 61, 61	0
mt8, mt6, mt4	4B / original, pressure	3×270	61, 61, 61	0
v2, v2b	4B / pilot pools	2×30	16, 16	0
q17_g600, q17_mt6	1.7B / original	2×270	32, 32	0
q8_g600, q8_mt6, q8_mt4	8B / original	3×270	78, 78, 78	0
v4_g600, v4_mt6	8B / hardened v4 (post-fix)	2×216	—	0
v5_g600, v5_mt6	8B / v5 pool (post-fix)	2×180	—	0
v5cur_g600, v5cur_mt6	8B / curated union	—	66, 66	0
q4cur_g600, q4cur_mt6	4B / curated (symmetric control)	2×198	53, 53	0
mbpp_g600, mbpp_mt6	4B / MBPP+	2×540	107, 107	0
mbpp17_g600, mbpp17_mt6	1.7B / MBPP+	2×540	78, 78	0
Canonical total		6,378	1,307	0
<i>Excluded from the canonical total:</i>				
v4_g600, v4_mt6 (pre-fix)	8B / hardened v4, ASLR leak	2×216	—	2
g600 (APC)	4B / original, APC on	270	68	7 + 2

Table 5: Run-by-run null-replay reconciliation. Canonical total $6,378 + 1,307 = 7,685$, all exact; including the pre-fix runs, 2/8,117 nulls ever run under the stated premises were inexact (both in one pre-fix run, §4.1). “Attached” nulls are the queue-floor checks run inside each interaction experiment.